Unit 2 – Selection and Iteration Name: **KEY**
HW 2

1) Consider the following code segment.

```
int x = 4;
int y = 5;
int z = 3;
```

Determine the output of the code below.

a)
```
if(x > 4) {
    if(y < z) {
        System.out.print("A");
    }
    else {
        System.out.print("B");
    }
}
else {
    if(y > z) {
        System.out.print("C");
    }
    else {
        System.out.print("D");
    }
}
```

C

b)
```
if((x + y) % z == 0) {
    if(x - z < 0) {
        System.out.print("A");
    }
    else {
        System.out.print("B");
    }
}
else {
    if((x + y) % z == 1) {
        System.out.print("C");
    }
    else {
        System.out.print("D");
    }
}
```

B

c)
```
if(y / 4 <= 1) {
    if(x + y + z != 15) {
        System.out.print("A");
    }
    else {
        System.out.print("B");
    }
}
else {
    if(z % 2 > 1) {
        System.out.print("C");
    }
    else {
        System.out.print("D");
    }
}
```

A

d)
```
if(y % z > 2) {
    if(x * y > 20) {
        System.out.print("A");
    }
    else {
        System.out.print("B");
    }
}
else {
    if(x * z != 12) {
        System.out.print("C");
    }
    else {
        System.out.print("D");
    }
}
```

D

2) Consider the following code.

```
if(temperature > 50) {
     if(raining) {
          System.out.println("No jacket, umbrella");
     }
     else {
          System.out.println("No jacket, No umbrella");
     }
}
else if(temperature < 40) {
     if(raining) {
          System.out.println("Jacket, umbrella");
     }
     else {
          System.out.println("Jacket, no umbrella");
     }
}
```

What are some example values of temperature and raining in order to print the following?

a) `Jacket, no umbrella` temperature less than 40 raining = false	b) `No jacket, umbrella` temperature greater than 50 raining = true	c) `Jacket, umbrella` temperature less than 40 raining = true
d) `No jacket, No umbrella` temperature greater than 50 raining = false	e) What values will nothing get printed? when temperature is any value 40 to 50	

A 3) Consider the following code.

```
if(x < 5) {
     if(y > 3) {
          System.out.println("First");
     }
     else {
          System.out.println("Second");
     }
}
else {
     if(y == 4) {
          System.out.println("Third");
     }
     else {
          System.out.println("Fourth");
     }
}
```

What will the output be if `x` is 3 and `y` is 4?

(A) First
(B) Second
(C) Third
(D) Fourth

4) The variable age contains the age of a person trying to buy a movie ticket. A ticket price is assigned as follows:

- If the person is under 18:
  - If they are under 12, print "$5 Ticket"
  - Otherwise, print "$7 Ticket"
- Otherwise (if the person is 18 or older):
  - If they are 65 or older, print "$5 Ticket"
  - Otherwise, print "$9 Ticket"

Complete the following code segment using nested if statements.

```
int age = /* not shown */ ;

if (age < 18) {
     if (age < 12) {
          System.out.println("$5 Ticket");
     }
     else {
          System.out.println("$7 Ticket");
     }
}
else {
     if (age >= 65) {
          System.out.println("$5 Ticket");
     }
     else {
          System.out.println("$9 Ticket");
     }
}
```